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CSA0910 – PROGRAMMING IN JAVA

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

```
import java.util.ArrayList;

public class RemoveDuplicates {

    public static void main(String[] args) {

        ArrayList<Integer> list = new ArrayList<>();

        int[] arr = {10, 20, 30, 10, 40, 20, 50, 60, 30, 70, 80, 50, 90, 100, 70}; for
        (int i = 0; i < arr.length; i++)

            list.add(arr[i]);

        for (int i = 0; i < list.size(); i++) {
            for (int j = i + 1; j < list.size(); j++) {
                if (list.get(i).equals(list.get(j))) {
                    list.remove(j);
                    j--;
                }
            }
        }

        System.out.println("ArrayList after removing duplicates:");
        System.out.println(list);
    }
}
```

```
RemoveDuplicates.java
1 import java.util.ArrayList;
2 public class RemoveDuplicates {
3     public static void main(String[] args) {
4         ArrayList<Integer> list = new ArrayList<>();
5         int[] arr = {10, 20, 30, 10, 40, 20, 50, 40, 30, 70, 80, 90, 90, 100, 70};
6         for (int i = 0; i < arr.length; i++) {
7             list.add(arr[i]);
8             for (int k = 0; k < list.size(); k++) {
9                 for (int j = k + 1; j < list.size(); j++) {
10                     if (list.get(k).equals(list.get(j))) {
11                         list.remove(j);
12                         j--;
13                     }
14                 }
15             }
16         }
17         System.out.println("ArrayList after removing duplicates:");
18         System.out.println(list);
19     }
20 }
```

ArrayList after removing duplicates:
[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

Code Execution Successful

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

```
import java.util.LinkedList;

public class EmployeeList {

    public static void main(String[] args) {

        LinkedList<Integer> emp = new LinkedList<>();

        emp.add(101);
        emp.add(102);
        emp.add(103);
        emp.add(104);
        emp.add(105);

        System.out.println("Forward Order:");

        for (int id : emp)

            System.out.print(id + " ");

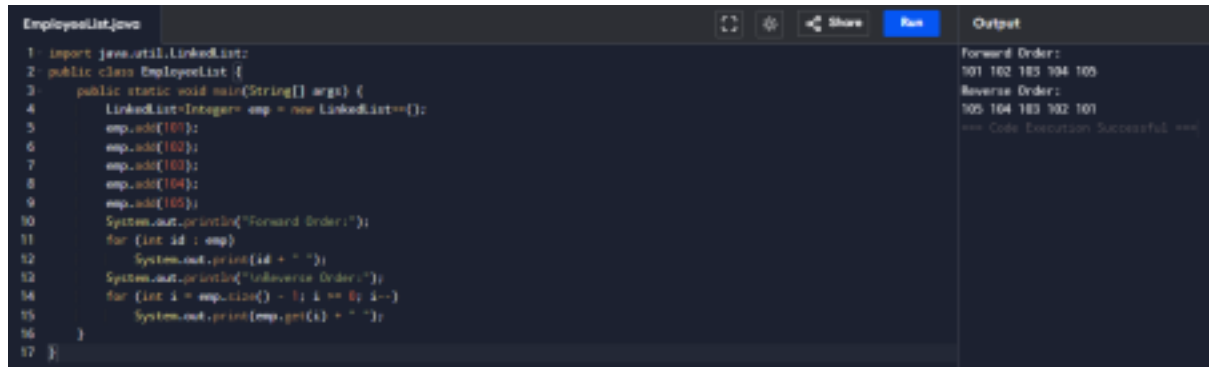
        System.out.println("\nReverse Order:");

        for (int i = emp.size() - 1; i >= 0; i--)

            System.out.print(emp.get(i) + " ");

    }

}
```

A screenshot of a Java IDE window titled 'EmployeeList.java'. The code defines a class 'EmployeeList' with a 'main' method that creates a 'LinkedList<Integer>' and adds five integers: 101, 102, 103, 104, and 105. It then prints the 'Forward Order' and 'Reverse Order' of the list. The 'Output' pane on the right shows the execution results: 'Forward Order: 101 102 103 104 105' and 'Reverse Order: 105 104 103 102 101'. A status bar at the bottom of the output pane indicates 'Code Execution Successful'.

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

```
import java.util.HashSet;

import java.util.Scanner;

public class UniqueEmails {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        HashSet<String> emails = new HashSet<>();

        System.out.print("Enter number of email IDs: ");

        int n = sc.nextInt();

        sc.nextLine();

        for (int i = 1; i <= n; i++) {

            System.out.print("Enter Email " + i + ": ");

            emails.add(sc.nextLine());

        }

        System.out.println("\nUnique Email IDs:");

        for (String email : emails)

            System.out.println(email);

        System.out.println("Total Unique Emails: " + emails.size());

    }

}
```

```

1: import java.util.HashSet;
2: import java.util.Scanner;
3: public class UniqueEmails {
4:     public static void main(String[] args) {
5:         Scanner sc = new Scanner(System.in);
6:         HashSet<String> emails = new HashSet<>();
7:         System.out.print("Enter number of email IDs: ");
8:         int n = sc.nextInt();
9:         sc.nextLine();
10:        for (int i = 1; i <= n; i++) {
11:            System.out.print("Enter Email " + i + ": ");
12:            emails.add(sc.nextLine());
13:        }
14:        System.out.println("\nUnique Email IDs:");
15:        for (String email : emails)
16:            System.out.println(email);
17:        System.out.println("Total Unique Emails: " + emails.size());
18:    }
19: }

```

Output

```

Enter number of email IDs: 2
Enter Email 1: 182424473.vinay@uavertha.com
Enter Email 2: rahulrsk78@gmail.com

Unique Email IDs:
rahulrsk78@gmail.com
182424473.vinay@uavertha.com
Total Unique Emails: 2

--- Code Execution Successful ---

```

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

```

import java.util.HashMap;

public class WordFrequency {

    public static void main(String[] args) {

        String text = "Java is easy Java is powerful Java language";

        String[] words = text.toLowerCase().split(" ");
        HashMap<String, Integer> map = new HashMap<>();

        for (String word : words) {

            if (map.containsKey(word))

                map.put(word, map.get(word) + 1);

            else

                map.put(word, 1);

        }

        System.out.println("Word Frequency:");

        for (String word : map.keySet())

            System.out.println(word + " : " + map.get(word));

    }

}

```

WordFrequency.java	Run	Output
<pre>1 import java.util.HashMap; 2 public class WordFrequency { 3 public static void main(String[] args) { 4 String text = "Java is easy Java is powerful Java language"; 5 String[] words = text.toLowerCase().split(" "); 6 HashMap<String, Integer> map = new HashMap<>(); 7 for (String word : words) { 8 if (map.containsKey(word)) 9 map.put(word, map.get(word) + 1); 10 else 11 map.put(word, 1); 12 } 13 System.out.println("Word Frequency:"); 14 for (String word : map.keySet()) 15 System.out.println(word + " : " + map.get(word)); 16 } 17 }</pre>	Run	<pre>Word Frequency: java : 3 powerful : 1 is : 2 language : 1 easy : 1 === Code Execution Successful ===</pre>